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Investigation of the Effects of Distributed Surface Roughness on Supersonic Flows

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Supersonic and hypersonic aircraft operate in high enthalpy environments that require the use of thermal protection systems (TPS). These TPS often protect vehicles from extreme heating loads via ablation, a process that involves coupled heat and mass transfer that removes material from a surface through aerodynamic heating. The removal of TPS material can cause cross-hatching and shape change; ultimately generating non-uniform roughness on the surface of a vehicle. These surface features caused by ablation can interact with the boundary layer and can lead to flow disturbances like acoustic waves, premature turbulent transition, and elevated heating rates. To investigate the possible effect of ablation-induced roughness on the surface of a supersonic vehicle, experiments were performed using PIV and Schlieren to quantitatively and qualitatively characterize the aerodynamic effects caused by distributed uniform and non-uniform surface roughness on a supersonic boundary layer. These experiments were performed at the Mach 2 wind tunnel facility at The University of Tennessee Space Institute (M = 2.01, $Re_x = 30 \times 10^6$ m^{-1}). Uniformly distributed compressibility effects were observed using both PIV and Schlieren for a diamond roughness model. Non-uniform waves were generated with the random roughness model and could also be identified with the diagnostic methods employed. For the diamond roughness and random roughness cases, the mean velocity profiles followed similar trends to the flat plate case. Reynolds stresses were also calculated, in which the diamond roughness had the highest Reynolds shear stress value near the wall, and the random roughness had the lowest.

Nomenclature

B	constant value for log-law equation
k	roughness height, m
k^+	non-dimensionalized inner-scaled roughness height
u_τ	friction velocity, m/s
u^+	normalized freestream velocity
U	streamwise velocity, m/s
U_o	freestream streamwise velocity, m/s
U_e	boundary layer edge velocity, m/s
V	wall-normal velocity, m/s
y	height above test section floor, m
y^+	normalized wall distance
δ	boundary layer thickness, m

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κ von Kármán's constant
ρ freestream density, kg/m^3
τ_w wall shear, Pa
ν_w kinematic viscosity at the wall, m^2/s

I. Introduction

Vehicles entering supersonic and hypersonic environments encounter high enthalpy conditions that require the use of thermal protection systems (TPS). By design, the extreme enthalpy conditions on the surface of the vehicle cause these TPS materials to ablate. Ablation is a process in which sacrificial surface material is slowly burned away to prevent the vehicle from overheating. The removal of this TPS material can cause various shape changes on the exposed surface; ultimately leading to non-uniform surface roughness, cross-hatching, and striations.[1] The effects of surface roughness and their impact on the surrounding flow need to be fully understood to prevent design flaws in high-speed aircraft.

The effects of both singular and distributed surface roughness elements on low speed flows have been extensively studied and well documented.[2-5] Wu et al.[2,3,5] collected high resolution PIV data over roughness elements replicated from surface scans of damaged turbine blades at conditions around 16 m/s. The topography of the roughness elements was highly irregular, which contained large-scale and small-scale representations of spallation, pitting, and surface deposition.[2,3,5] Large increases in Reynolds stresses in regions of large surface defects at lower speed regimes were reported.[2,3,5] Future efforts by the HORIZON group at UTSI will expand the characterization of the flow over these roughness topographies into supersonic flows.

Latin and Bowersox[6,7] collected Schlieren, laser Doppler velocimetry, and pitot probe data at Mach 2.9 over sandpaper, a 3D machined roughness element, and a 2D machined roughness element to measure and characterize the mean flow properties and energy spectra. These studies described the effects of various roughness topography elements on the magnitude of the mean and turbulent statistical properties on a supersonic boundary layer. The studies also investigated various scaling processes for the mean and turbulent properties. Peltier et al.[8] investigated crosshatch roughness distortions on a hypersonic turbulent boundary layer using diamond-shaped crosshatching and imaged the flow with Schlieren and Particle Image Velocimetry (PIV). This study showed an oscillatory response of the mean flow to shock waves and expansion fans that were generated by distributed diamond roughness elements.[8] It was also determined that the diamond roughness elements were responsible for increased turbulent fluctuations in the lower half of the boundary layer, while also decreasing turbulent stresses in the upper half of the boundary layer.[8] Sahoo et al.[9] similarly collected data on the behavior of turbulence with complex boundary conditions and surface roughness and noted significant compressibility effects associated with roughness elements at high Mach numbers (M=7.3).

The objective of this study is to study the effects of uniform and non-uniform distributed surface roughness on supersonic boundary layers using optical diagnostics. Three test models were used: a flat plate floor used to characterize the undisturbed supersonic boundary layer, a diamond roughness floor insert to determine the boundary layer effects from post-ablated surfaces, and a randomly distributed roughness floor insert. The results of the diamond roughness elements are compared to Peltier et al.[8] and Sahoo et al.,[9] since the geometries are similar. The impact of each roughness element on the supersonic boundary layer was characterized with Schlieren imaging and PIV. The roughness elements investigated in the current work were inspired by pre- and post-test arc-jet TPS test coupons, and a primary goal of this research effort is to develop an improved understanding of the effect of ablation-induced roughness on arc-jet tests in an environment where the aerodynamic effects have been decoupled from the complex high-enthalpy flow chemistry present in arc-heated facilities.

II. Experimental Facilities

A. Experimental Facility

The wind tunnel used for these experiments was the Mach 2 low-enthalpy blowdown wind tunnel facility located at The University of Tennessee Space Institute (UTSI). The Mach 2 wind tunnel has a constant test section cross section of 203 mm × 203 mm (8" × 8"). This wind tunnel operates with air supplied by a 23.6 m^3 bottle farm with a maximum pressure of 20 MPa. The stilling chamber pressure is maintained at 210 kPa by an upstream control valve and the maximum run duration is approximately two minutes. A schematic of the Mach 2 wind tunnel can be seen in Figure 1. The airflow is not heated and the stagnation temperature is approximately 285 K.[12,15] The average freestream velocity and Mach number are 507 m/s and 2.01 respectively.[12,15] The expected unit Reynolds number is approximately 30×10^6 m^{-1}.[12,15] At the location chosen for the experiments, a fully-developed 11 mm turbulent boundary layer is present on the test section floor. The test section has optical access on the tunnel sidewalls and

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ceiling provided by BK7 glass windows. Also, the test section floor allows for interchangeable floor inserts, making the facility suitable for testing with multiple interchangeable roughness elements.

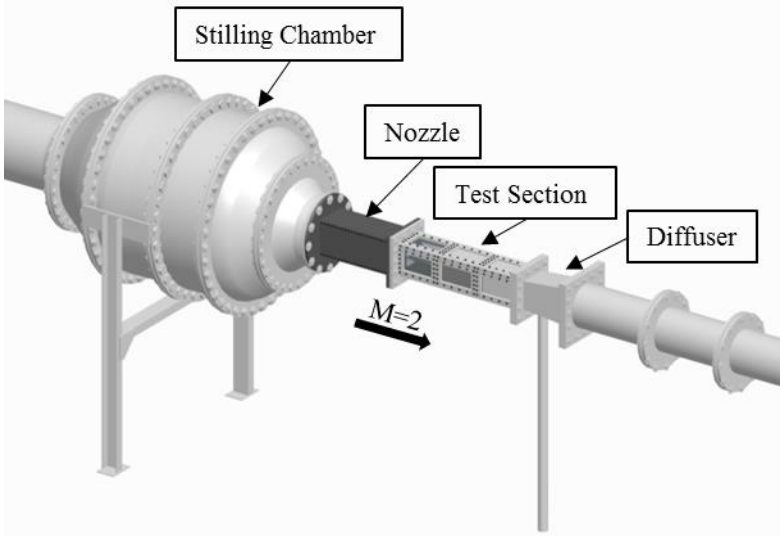

Figure 1. Schematic of the UTSI high-speed wind tunnel.

B. Model Geometries

Three models were designed and tested in the study. The first model was a flat plate floor insert machined to be smooth and flush with the wind tunnel floor (Figure 2(a)), the second model was a flat floor insert with an array of uniformly distributed diamond roughness elements (Figure 2(b)), and the third model was a flat floor insert with random roughness distributed in a non-uniform manner (Figure 2(c)). All models were fabricated from 12.7 mm (0.5") thick clear acrylic to allow the laser sheet to pass through the surface; therefore, preventing laser reflections in the near-wall region. All roughness models were inserted into the test section floor to align the top of the roughness element flush with the floor of the upstream smooth surface, as done in similar previous experiments performed by Latin et al.,[6,7] Peltier et al.,[8] and Sahoo et al.[9] Test article parameters can be found in Table 1.

The diamond roughness elements were sized to match the results found from both the experimental campaigns of Peltier et al.[8] and Sahoo et al.[9] The scaling parameter used for comparison was the non-dimensionalized inner-scaled roughness height, k^+, which can be written as,

$$k^+ = \frac{k u_\tau}{\nu_w} \tag{1}$$

where k is the roughness height, u_τ is the friction velocity, and ν_w is the viscosity at the wall. The friction velocity can be written as,

$$u_\tau = \sqrt{\frac{\tau_\mathrm{w}}{\rho}} \tag{2}$$

where τ_w is the shear stress at the wall and ρ is the freestream density.

Matching the k^+ value should permit a comparison of the results of the current work to previous experiments despite the varying freestream conditions. One of the variables used to calculate the k^+ value is the friction velocity, u_τ. Peltier et al.[8] and Sahoo et al.[9] used the Clauser chart method with Van Driest compressibility scaling[10] to estimate friction velocity of 69 m/s and 143 m/s for their diamond roughness models. This study, using the Clauser chart method as well, calculated a friction velocity of 29.28 m/s for the diamond roughness model. Wei et al.[11] discussed the difficulties and concerns of using the Clauser chart method for calculating the friction velocity, which could ultimately lead to possible uncertainties in the friction velocity. The diamond roughness friction velocity was then used to calculate a k+ value of 270. This k+ value was different than the original design k+ value of 155, which was based on an anticipated friction velocity. The diamond element width and length for this experiment were derived from using a standard end mill bit size of 3.8 mm (0.15"), while keeping a diamond roughness element in the exact center of the model. The resulting channels between the diamond elements were square, matching the geometries of Sahoo et al.,[9] rather than the hemispherical channels employed by Peltier et al.[8] The roughness element height was 0.33 mm (0.013").

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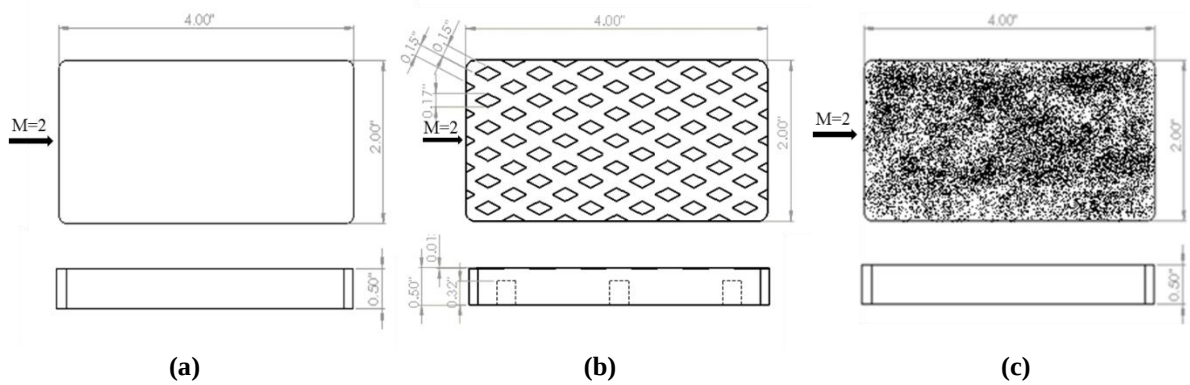
The random roughness model was created by randomly translating square and ball end mill bits across an acrylic insert. The resulting floor insert was sand blasted to remove any smooth edges. The maximum depth of any roughness created by the end mill or sand blasting were limited to depths close to the diamond roughness elements. Exact depths of the roughness elements were not determined for this effort, but the groove depths ranged from 0.25 to 0.4 mm. This floor insert is a first-order proof-of-concept effort and future experiments will use random roughness models similar to those employed in Wu et al.[5]



(a) (b) (c)

Figure 2. (a) flat floor insert, (b) diamond roughness floor insert, and (c) random roughness floor insert.

Table 1. Test articles parameters for Peltier et al.,[8] Sahoo et al.,[9] and the current experiments.

Roughness Type	M (--)	k (mm)	k^+ (--)	u_τ (m/s)
Flat Plate	2.01	0	0	30.23
Diamond Roughness	2.01	0.33	270	29.28
Random Roughness	2.01	0.25 to 0.4	202 to 404	29.76
Peltier et al.[8] Diamond Roughness	4.9	3.3	160	69
Sahoo et al.[9] Diamond Roughness	7.3	1.65	154	143

C. Schlieren Experimental Setup

Schlieren imaging was a secondary, qualitative imaging technique used for flow visualization of the fluid structures interacting with the distributed roughness models. The Schlieren system at UTSI consisted of a Z-type optical setup with 2.67 m focal length mirrors, a high-speed LED light source, and a Photron FASTCAM MINI UX100 high-speed camera. For these experiments, the Photron camera acquired images at a frame rate of 50 Hz (1,280 × 1,024 pixels up to 4 kHz). The light source was a custom-built high-repetition rate LED light source with a pulse duration of approximately 700 ns, fast enough to effectively freeze the flow for the resolution employed in the current work.

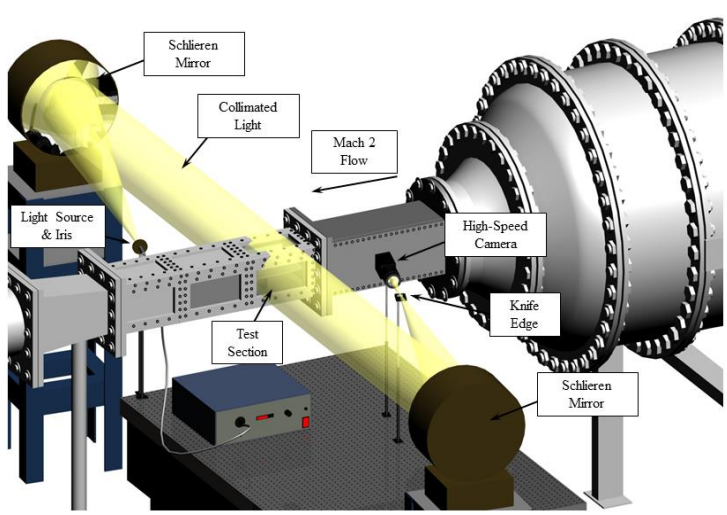


Figure 3. Illustration of the Z-type Schlieren system.[12]

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D. Particle Image Velocimetry Experimental Setup

Two-component Particle Image Velocimetry (PIV) was the primary optical diagnostic employed to quantitatively characterize fluid flow over the test articles. Degussa P25 Aeroxide Titanium Dioxide (TiO_2) was used as the solid seeding particle, with a nominal particle diameter of 21 nm.[13] A particle lag analysis was completed following the process outlined in Hou[13] to calculate a particle Stokes number of 0.09 for the Mach 2 boundary layer conditions in the current work. This result satisfies the guideline for reliable flow tracking established by Samimy and Lele,[14] which requires a particle Stokes number of less than 0.5. The particles were injected directly into the Mach 2 facility, upstream of the convergent section of the wind tunnel nozzle via a single-stage fluidized bed driven by compressed air. The seeder used a 12.7 mm diameter injector rake in the shape of a straight cylinder with a series of 2 mm diameter holes. The rake holes were oriented in the upstream direction, and in the same plane as the laser sheet to promote a uniform distribution of the particles over the domain of interest. Only the bottom two holes were open to promote optimal seeding in the boundary layer.

The seed particles were illuminated by two sheets of 532 nm light from a frequency-doubled Nd:YAG laser (Continuum Surelite PIV I-10) and imaged using a CCD camera (IMPERX B3340, 3312 × 2488 pixels, 0.011 mm/px), as seen in Figure 4. The laser light was delivered through a window on the ceiling of the test section using laser mirrors. The light was formed into a thin (< 1 mm thick) sheet by passing the beam through a 400 mm spherical and -50 mm cylindrical lens before transmission through the test section. The corresponding laser sheet was aligned with the freestream and on the centerline of the test model. The CCD camera was oriented perpendicular to the laser sheet and the laser pulses were separated by 500 ns to allow for a sufficient pixel shift in the particle image pairs.

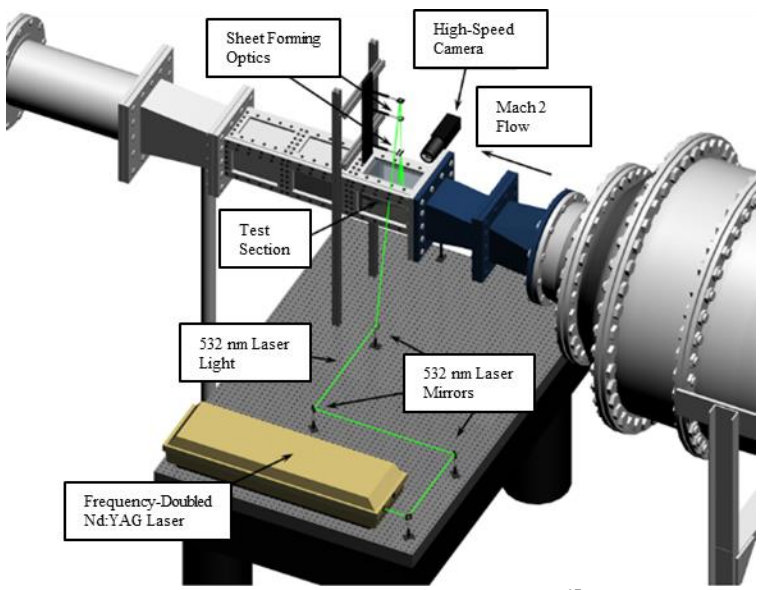

Figure 4. PIV experimental setup.[15]

III. Results

A. Schlieren Results

Schlieren images were collected to qualitatively analyze the boundary layer profiles for all three models. All Schlieren images shown in this manuscript were acquired at 50 Hz, with a horizontal knife-edge to show the strong density gradients in the wall-normal direction. The freestream Mach number was 2.01.[12,15] Figure 5 shows a full field-of-view instantaneous Schlieren image of the air flow over the flat plate model. The red rectangle represents the field-of-view in all three images in Figure 6, the blue overlaid line corresponds to the location of the floor inserts, and the yellow dotted line corresponds to the field-of-view of the PIV camera. There is a weak shock wave at an angle of 32° relative to the test section floor located at the leading edge of all three plates generated from a small step at the interface of the test section floor and the flat plate model. At Mach 2, Mach waves would be at 30º relative to the test section floor. In Figure 5, there is a fully developed turbulent boundary layer visible on the floor of the wind tunnel and over the flat plate model, where turbulent structures in the boundary layer are apparent. The other waves in the image are

generated from the junction of the nozzle and test section as well as the leading edges of the primary floor and ceiling inserts.

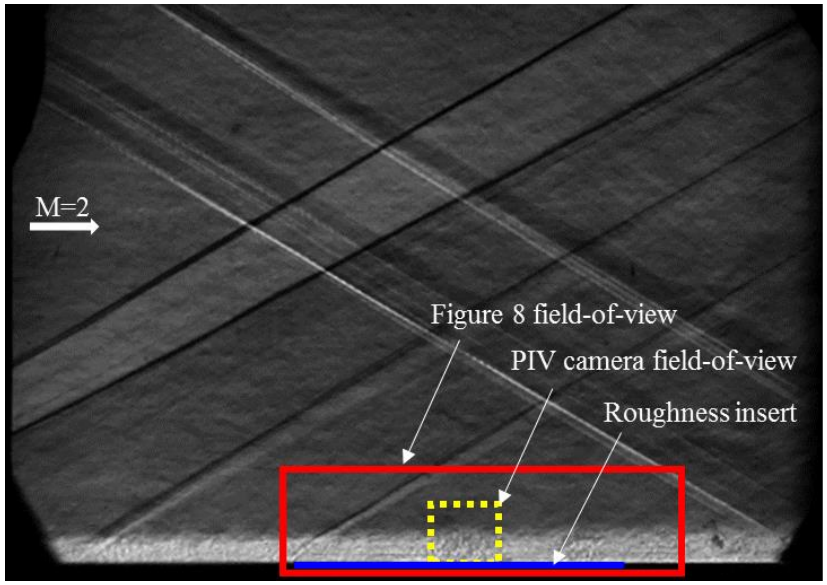

Figure 5. Full field-of-view Schlieren image over the flat plate model.

Figure 6(a) shows an instantaneous Schlieren image over the flat plate model. This is a cropped section from Figure 5 to show only the regions of interest for this study. The only major compressibility effect caused by the flat plate is a leading edge shock at the beginning of the model as the incoming turbulent boundary layer appears to be otherwise unperturbed. The lack of perturbations in the boundary layer will provide a good baseline result for comparison to the other roughness models.

Figure 6(b) shows an instantaneous Schlieren image of the flow over the diamond roughness element. There are strong gradients – the diagonal black lines – distributed across the entire diamond roughness model. These black lines are at 33º relative to the test section floor, which indicates they are shock waves and not Mach waves. These shocks originate from the sharp leading edge of the distributed roughness elements. It is important to note that Schlieren imaging is path integrated, therefore the shocks seen in this figure are multiple shocks along the light path that are averaged together. Directly behind each shock wave indicates the presence of expansion waves. These can be seen as the regions just behind the shock waves that are brighter than the freestream. The trends seen in Figure 6(b) closely resemble the Schlieren results from Peltier et al.[8] The only noticeable difference is the angle at which the shock waves form, which is to be expected because of the difference in the freestream Mach number. An interesting feature that occurs in both studies is that the shock waves travel through the entire boundary layer and into the freestream. Therefore, the boundary layer and freestream conditions are affected by the compressibility effects caused by the distributed diamond roughness elements. Also, there is a shock wave that occurs at the trailing edge of the model due to a 0.33 mm step from the floor of the roughness insert to the wind tunnel floor. The shock waves should cause local regions of increased wall-normal velocities which were measured by PIV and shown in Figure 9.

Figure 6(c) is an instantaneous image of the flow over the random roughness model. Since the roughness is randomly distributed over the length of the test article, there are fewer spatially averaged features compared with the diamond roughness model. This leads to fainter flow features in the field of view for the Schlieren images. However, the flow is still notably different from the flat floor model, with waves clearly visualized along the length of the plate. A majority of the compressibility effects imaged in Figure 6(c) were measured at 30°, which would indicate that they are mostly Mach waves, rather than shock waves. These Mach waves travel through the entire boundary layer and into the freestream. From these Mach waves, there should be an increase in the wall-normal velocity fluctuations as compared to the flat plate case, which was verified by PIV and shown in Figure 10.

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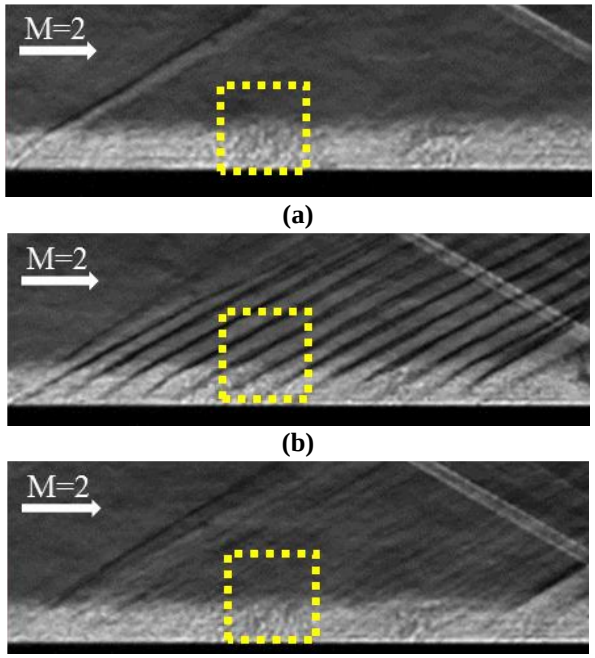

Figure 6. Schlieren images for the (a) flat plate insert, (b) diamond roughness insert, and (c) random roughness insert, recorded at 50 Hz, where the yellow dashed line represents the PIV field-of-view.

B. 2D Centerline PIV

1. PIV Post Processing

PIV was employed at the centerline of the test section to characterize the turbulent boundary layer and the flow features resulting from the roughness elements. The resulting raw PIV images were processed using LaVision's DaVis software. The PIV setup described in Section II.D was used to properly orient the laser sheet with respect to the freestream flow. Calibration images were captured before each day of testing to ensure consistency in the camera field-of-view by placing a ruler at the centerline of the test section, aligned with the leading edge of the roughness insert. For the LaVision scaling procedure, common points were chosen along the ruler image to keep consistency for all PIV images. The ruler images from each test were also used to correct for any potential camera rotation and verify the magnification of the images.

Once the raw images were scaled and rotated properly, LaVision's DaVis software was used to process the images. The software began calculating instantaneous vector fields using a two-pass, 2:1 elliptic 128 × 128-pixel interrogation window with 50% overlap. A second vector field calculation iteration used a three-pass adaptive 48 × 48-pixel interrogation window with a 50% overlap, resulting in a 14.2 × 13.7 mm field-of-view, interrogation windows of 0.5 × 0.5 mm and final vector fields of 54 × 52 vectors. The raw vector fields were then imported into MATLAB for post-processing. The origin for the analysis was defined as the leading edge of the roughness element, therefore, the measurements were collected at a location starting at 3.7 x/δ. A filter was applied to identify all velocity vector fields with greater than 90% valid vectors, remove any outlier data points, and then a nearest neighbor linear interpolation technique was used to fill in any missing vectors. After processing all captured images, the statistical convergence of the mean streamwise velocity, the r.m.s. velocity profiles, and the Reynolds stresses were investigated using the process outlined by Combs.[16] The mean, r.m.s. and Reynolds stress data for all roughness models converged to within 0.1%, which was assumed to be sufficiently converged for the analysis performed in the current work.

2. Instantaneous PIV Vector Fields

Instantaneous velocity fields are shown in Figure 7(a), Figure 7(b), and Figure 7(c) for the flat plate, diamond roughness, and random roughness models. The local instantaneous velocity fluctuations are represented as arrows in the figures. These figures are representative examples for images that have at least 90% valid vectors. All three figures show that the boundary layers over all roughness models are turbulent. The size and strength of the turbulent structures vary in an instantaneous sense. The figures show that it is difficult to determine any common flow patterns from the instantaneous velocity vector fields.

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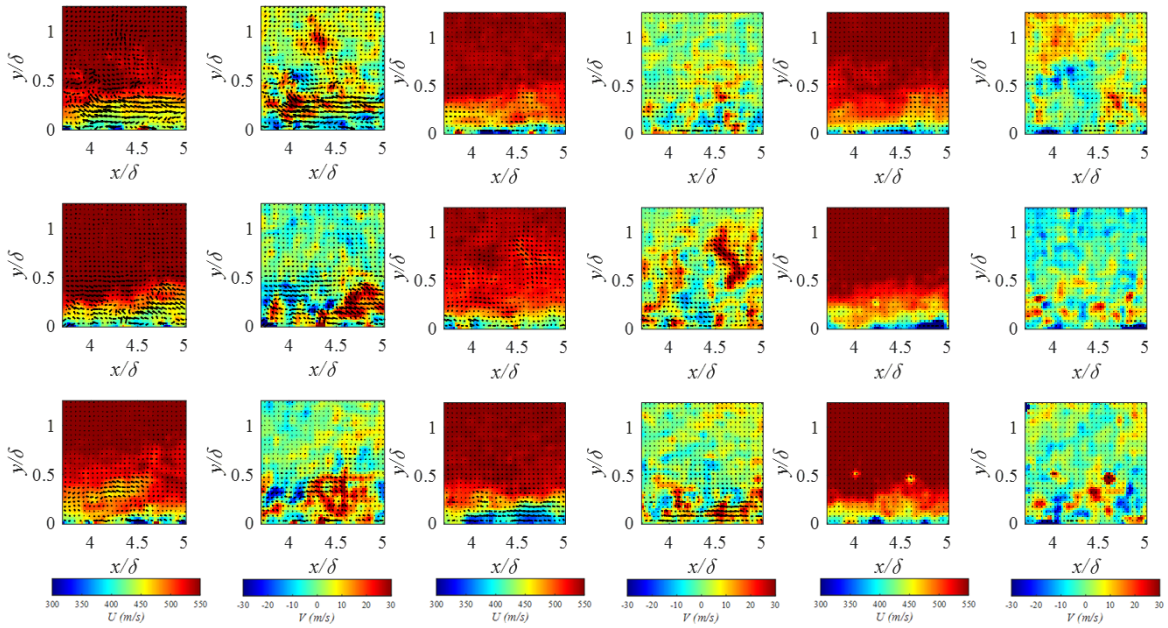

(a) **(b)** **(c)**

Figure 7. Instantaneous velocity fields at different times for the (a) flat plate, (b) diamond roughness, and (c) random roughness floor inserts; where the arrows represent local velocity fluctuations.

3. Mean Vector Fields

Figure 8 shows the two-dimensional mean streamwise and wall-normal velocity fields for the flat plate model averaged from 384 vector fields with valid vectors above 90% collected during 9 separate runs. The arrows in Figure 8, Figure 9, and Figure 10 represent the local mean velocities, where the streamwise velocity was scaled by a factor of 50. A scaling was needed on the streamwise mean velocities since the wall-normal velocities were an order of magnitude lower. The boundary layer flow over the flat plate shows a dominant streamwise velocity component, with mean wall-normal velocities of less than 0.1% of the freestream velocity measured throughout the boundary layer, as expected for a flat plate. The velocity at the edge of the boundary layer for the flat plate insert was 501 m/s, which agrees with the reported conditions for the undisturbed test section boundary layer within 1.2%.[12,15]

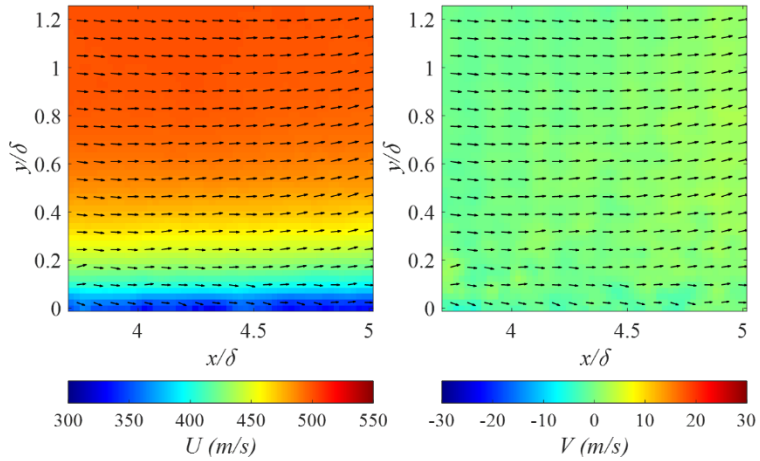

Figure 8. (left) Mean streamwise velocity field for the flat plate model with arrows representing local average velocities, with local streamwise velocities scaled by a factor of 50 and (right) mean wall-normal velocity field for the flat plate model with arrows representing local average velocities, with local streamwise velocities scaled by a factor of 50.

Figure 9 shows the two-dimensional mean streamwise and wall-normal velocity fields for the diamond roughness model averaged from 439 vector fields with valid vectors above 90% collected during 17 separate runs. In both the streamwise and wall-normal velocity fields, the effects of the distributed oblique shocks are apparent. The shock waves are located at the front of the yellow diagonal regions in the right image. Compressibility effects from oblique shock waves will deflect the flow upward, as seen in Figure 9. The wall normal velocities behind the oblique shocks increased from less than 1 m/s to 8-10 m/s. The flow then turns parallel to the test section floor, indicating the presence of expansions downstream of each oblique shock. These expansion fans can be seen in Figure 6(b) behind each shock wave. The velocity at a y/δ of 1 for the diamond roughness insert is 491 m/s, which is 10 m/s below the freestream velocity.

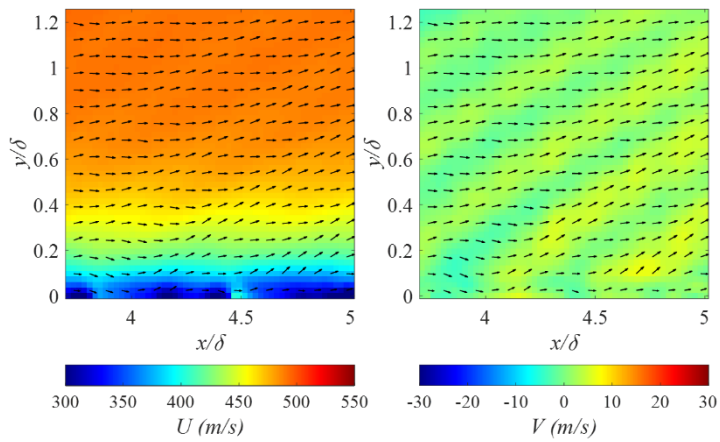

Figure 9. (left) Mean streamwise velocity field for the diamond roughness model with arrows representing local average velocities, with local streamwise velocities scaled by 50 and (right) mean wall-normal velocity field for the diamond roughness model with arrows representing local average velocities, with local streamwise velocities scaled by 50.

Figure 10 shows the two-dimensional mean streamwise and wall-normal velocity fields for the random roughness model averaged from 624 vector fields with valid vectors above 90%. The mean streamwise velocity field appears to be similar to the flat plate model, reaching a streamwise velocity of 496 at y/δ of 1. The magnitude of the wall-normal velocity field is higher than the flat plate model - approximately 5 m/s - due to the Mach waves. The Mach waves deflect the flow upward, but not as much as the diamond roughness model.

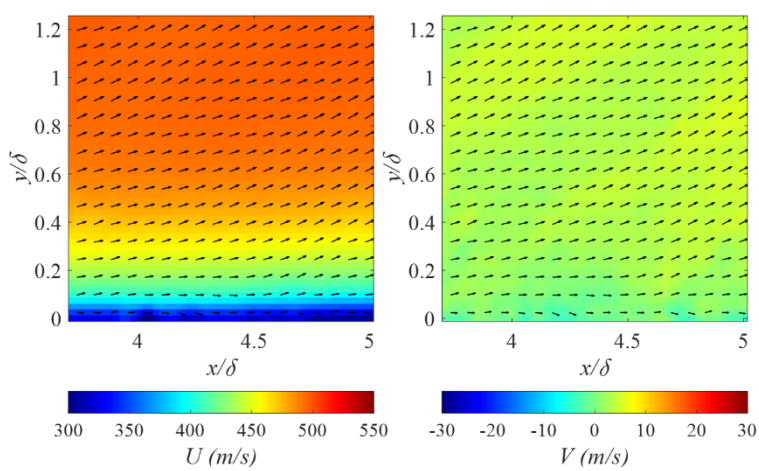

Figure 10. (left) Mean streamwise velocity field for the random roughness model with arrows representing local average velocities, with local streamwise velocities scaled by 50 and (right) mean wall-normal velocity field for the random roughness model with arrows representing local average velocities, with local streamwise velocities scaled by 50.

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4. Mean Flow Result

The mean velocity profiles, measured with PIV, for all three roughness models are shown in Figure 11. The PIV-measured streamwise velocity was normalized by the edge velocity of the boundary layer over each roughness element, while the height off the wall was normalized by the turbulent boundary layer height in the wind tunnel test section, δ. Peltier et al.[8] showed that the errors associated with averaging velocity values along the roughness elements in the x-direction were within the magnitude of the PIV uncertainty and it is therefore assumed that the effect of averaging in the x-direction is negligible. The three boundary layer profiles seem to be very similar, never varying by more than 1 to 3% for each y/δ location. This begins to show that in the region that PIV measurements were taken, there does not appear to be an indication of roughness effects.

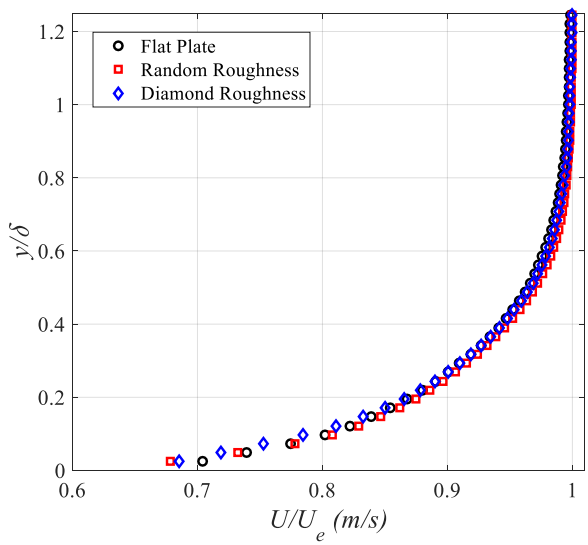

Figure 11. Boundary layer profile comparison between the flat plate, diamond roughness, and random roughness cases.

The velocity data were then transformed using the Van Driest transformation.[17] Since the mean velocity data are above a $y^+$of 30, the data collected should fall primarily within the log-law region. The log-law can be written as,[17]

$$\frac{U}{u_\tau} = u^+ = \frac{1}{\kappa}\ln y^+ + B \tag{3}$$

where κ, the von Kármán's constant, was set to 0.4, y^+ represents normalized wall units, u^+ represents normalized velocity, u_τ is the friction velocity, and B is a constant set to 5.1. Wall distances can be converted to normalized wall units using the following formula,[17]

$$y^+ = \frac{y u_\tau}{\nu_w} \tag{4}$$

where ν_w is the kinematic viscosity at the wall.

The data for the smooth plate are in good agreement with the log-law for y^+ values between 300 and 2,500, as seen in Figure 12. The flat plate model converges to a maximum u^+ value of 27.1. The flat plate case was then compared to both the diamond roughness case and the random roughness case. The diamond roughness model converges to a higher maximum u^+ value than the flat plate model at 27.7. The random roughness case converges to a maximum u^+ value of 27.33. This upward trend in u^+ values in the outer boundary layer for the diamond roughness model differed from the trends reported in Pelter et al.[8] and Sahoo et al.[9] Peltier et al.[8] (M = 4.9, δ = 10.9mm) reported a downward shift in u^+ of 13 from the flat plate model to the diamond roughness model, while Sahoo et al.[9] (M=7.3, δ=10mm) reported a downward shift of 7.3. Wu et al.[2] reported a downward shift of u^+ of 8.2 for a k^+ of 207 in low speed flow of 16.2 m/s. Again, the unexpected trend of the boundary layer could be attributed to the location at which the PIV measurements were taken. For instance, Peltier et al.[8] took PIV measurements at an x/δ location of 40, whereas this

research measured data at an x/δ location of 3.7. The trends witnessed in this experiment warrant future research efforts.

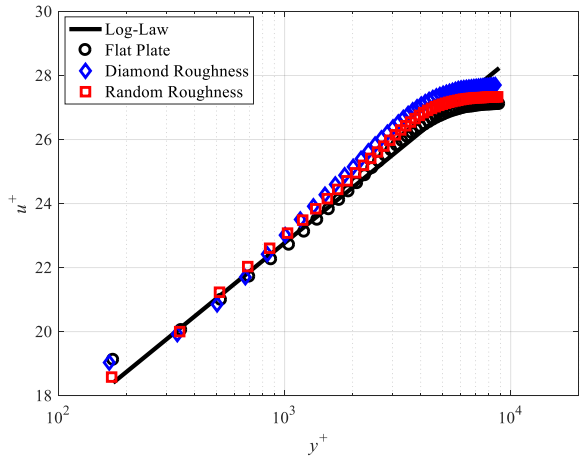

Figure 12. Van Driest corrected log-law plotted in normalized wall units for all three roughness models.

5. Reynolds Stresses

The Reynolds stresses for all three roughness elements scaled by the friction velocity for each roughness element are plotted in Figure 13. From the streamwise ($u'u'$) and wall-normal ($v'v'$) Reynolds stress terms, it is evident that a majority of the Reynolds stress term comes from the streamwise component, u', rather than the wall-normal direction, v'. This trend is expected since the streamwise velocity is, on average, a factor of 10 higher than the wall-normal velocity. Figure 13 shows that the diamond roughness model has a Reynolds stress that exceeds the flat plate in lower y/δ regions, and the random roughness had a lower Reynolds stress in lower y/δ regions. Peltier et al.[8] reported a differing trend with a reduction in magnitude in the near wall regions for roughness elements.

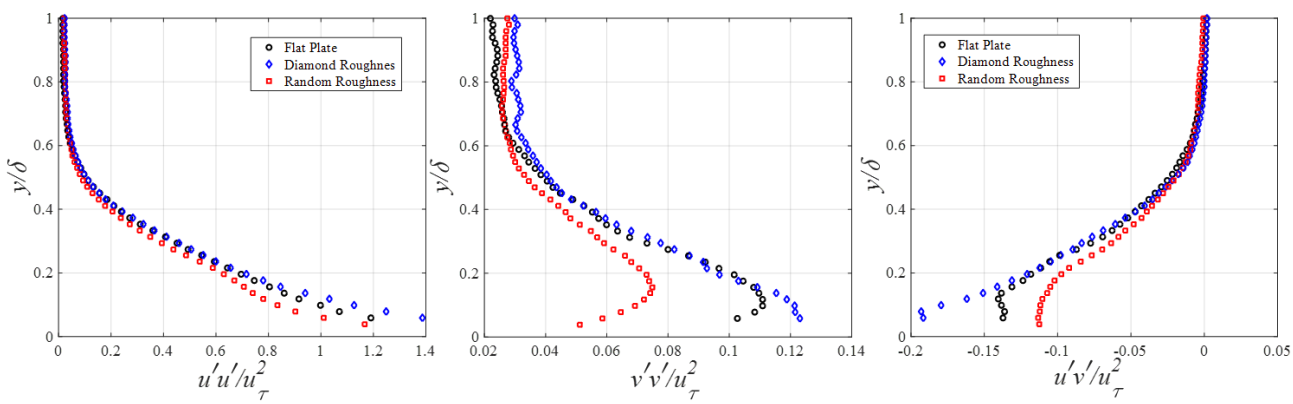

Figure 13. Reynolds Stresses, $u'u'$, $v'v'$, and $u'v'$, scaled by friction velocity, u_τ^2, for all three roughness models.

6. Probability Density Functions

Probability density functions (PDFs) are provided in Figure 14 for the $u'v'$ Reynolds stress term for each roughness model, normalized by the friction velocities. The PDFs were generated over a range of y/δ values to investigate the specific positive and negative velocity fluctuation contributions to the Reynolds stress terms. The negative skewness and mean trends of the near wall PDFs provide insight into the larger magnitude of Reynolds stress in the near wall region. Also noticeable is the larger probability for higher magnitudes of Reynolds stress in the near wall region. As y/δ increases, the probability of higher Reynolds stresses decreases, and the average value approaches zero. The trends seen in the PDF analysis are similar to what Peltier et al.[8] reported.

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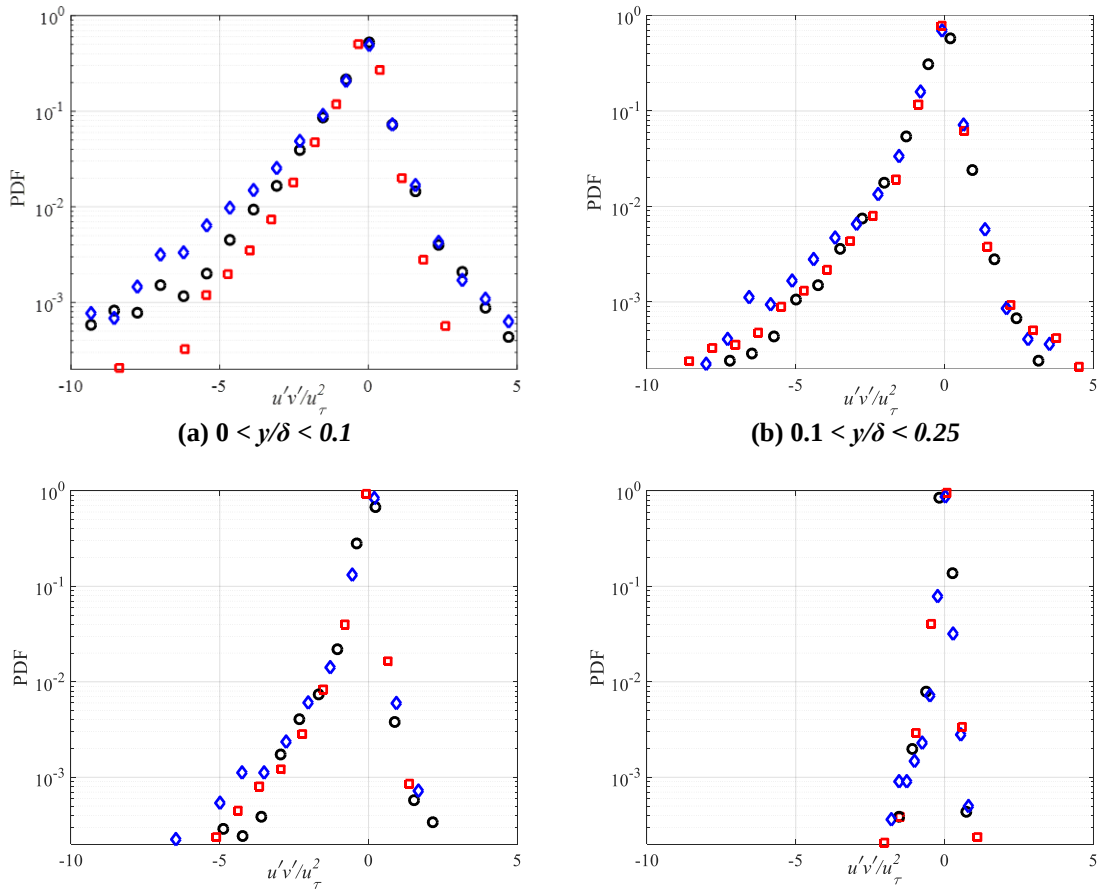

(a) 0 < *y/δ* < 0.1 **(b) 0.1 < *y/δ* < 0.25**

(c) 0.25 < *y/δ* < 0.5 **(c) 0.5 < *y/δ* < 1**

Figure 14. PDFs of the Reynolds Stress values for all three roughness elements, normalized by each respective friction velocity over a range of *y/δ*.

IV. Conclusion

PIV and Schlieren were employed in the UTSI Mach 2 wind tunnel (M = 2.01, $Re_x = 30 \times 10^6$ m^{-1}) to study the effects of floor-mounted distributed roughness elements on the supersonic boundary layer. Three test models were used in the experiments: one with no roughness, one with uniformly distributed diamond roughness elements, and one with non-uniform randomly distributed roughness. The k^+ value for the diamond roughness model was 270, and the random roughness model roughness height was between 0.25 to 0.4 mm. The random roughness model was a first-order proof-of-concept to study the flow effects seen from non-uniformly distributed roughness elements. Schlieren imaging for the flat plate floor insert showed a fully developed turbulent boundary layer, with a weak leading edge shock. The Schlieren image for the diamond roughness floor insert showed distributed oblique shocks and expansion waves along the entire duration of the test article. The Schlieren image of the random roughness floor insert showed that a majority of the compressibility flow features generated from the roughness element were Mach waves, rather than shock waves.

PIV instantaneous velocity vectors were calculated using LaVision's Davis software. Instantaneous PIV images showed varying sizes and strengths of turbulent structures. The resulting mean velocities for all three roughness models were plotted using a Van Driest transformation and compared against one another. The flat plate, diamond roughness and random roughness models agreed with the theoretical log-law curve between $300 < y^+ < 2500$, but varied in the maximum u^+ values (27.1 for the flat plate, 27.7 for the diamond roughness model, and 27.33 for the random roughness model). The trends seen in this analysis seemed to differ from the trends reported by both Peltier et al.[8] and Sahoo et al.,[9] who reported downward shifts in u^+ for the diamond roughness models versus the flat plate model of 7.3 and 13. This trend deviation will be investigated in future research. Reynolds stresses were calculated and scaled by the friction velocity for all three roughness models. The diamond roughness showed a slight increase in magnitude of the Reynolds stresses, scaled by friction velocity, in the near-wall region as compared to the flat plate

case. The random roughness showed a decrease in Reynolds stress as compared to the flat plate case near the wall. The addition of the PDFs validate the observation that the magnitude of Reynolds stress is highest in the near wall region for all roughness models.

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